🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.NOS.6

Graph on the Coordinate Plane Flagship

Lesson 9-1-flagship · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

LOST COVE MISSION

Captain Vega's Coordinates

You are the navigator on Captain Vega's expedition. Instead of 'X marks the spot,' she hid the treasure route as ordered pairs scrawled across the first quadrant of her chart, using only positive coordinates. Plot every point in the right order and the lost cove treasure is yours.

My Notes
Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can plot and identify points on the coordinate plane using ordered pairs.

Language Objective: I can explain my work using the words coordinate plane, ordered pair, origin, and quadrant.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Coordinate plane | Ordered pair | Origin | Quadrant | Reflection | Integer

👆 *Tap any word to see what it means and a picture.*

1. The grid made by a horizontal and a vertical number line is the ______________.

2. Two numbers (x, y) that tell the location of a point are an ______________.

3. The point (0, 0) where the x-axis and y-axis cross is the ______________.

4. One of the four regions of the coordinate plane is a ______________.

5. A flip of a point or shape over an axis is a ______________.

6. A whole number or its opposite, with no fraction part, is an ______________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Your map says the treasure is at (4, 2). Walk your partner through every move you make from the origin, using the words x-axis and y-axis?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Coordinate plane** — A grid with a line going across and a line going up that cross.
  Plano cartesiano — Una cuadrícula con una línea horizontal y una vertical que se cruzan.
- ☐ **Ordered pair** — Two numbers (x, y) that tell where a point is on a grid.
  Par ordenado — Dos números (x, y) que dicen dónde está un punto en una cuadrícula.
- ☐ **Origin** — The point (0, 0) where the two grid lines cross.
  Origen — El punto (0, 0) donde se cruzan las dos líneas de la cuadrícula.
- ☐ **Quadrant** — One of the four parts of a coordinate grid.
  Cuadrante — Una de las cuatro partes de una cuadrícula de coordenadas.
- ☐ **Reflection** — A flipped, mirror image across a line.
  Reflexión — Una imagen reflejada al otro lado de una línea.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I move ____ units along the x-axis.

Me muevo ____ unidades por el eje x.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We start at the origin and move along the x-axis first because ____.

Evidence: Student states the x-coordinate is the sideways move and the y-coordinate is the up/down move, both starting from the origin.

Reasoning: This evidence connects the definition of coordinate plane to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I move ____ units along the x-axis.**
  Me muevo ____ unidades por el eje x.
- **Then I move ____ units along the y-axis.**
  Luego me muevo ____ unidades por el eje y.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Coordinate plane
2. **Notes 2:** Ordered pair
3. **Notes 3:** Origin
4. **Notes 4:** Quadrant
5. **Notes 5:** Reflection
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Graph on the Coordinate Plane is swapping the x- and y-coordinates when plotting a point — moving up first using the y-value, then moving right using the x-value, instead of always reading the ordered pair as (horizontal, vertical). For the point (3, 5), this mistake lands the dot at (5, 3) instead of the correct spot: right 3, then up 5. Before you submit, retrace your steps and check that you moved right/left using the FIRST number and up/down using the SECOND number.*
8. **Try It 1:** A. Quadrant II — *The x-coordinate is negative (left) and the y-coordinate is positive (up). Negative x and positive y land in the top-left region, Quadrant II.*
9. **Try It 2:** A. On the y-axis at (0, 8) — *An x-coordinate of 0 means no horizontal move, so the point stays on the y-axis. Moving up 8 places it at (0, 8).*